
Subject Re-Identification Leakage in Motor-Imagery BCI Models

A Multi-Corpus, Audit-Clean Benchmark of Attacks and Defenses

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Abstract

Motor-imagery brain–computer interfaces (BCIs) are typically evaluated by their ability to decode neural activity into task commands. We evaluate a complementary privacy question: whether the EEG representations learned or exposed by such systems also encode stable subject-specific information that enables re-identification, verification, or membership inference. We present a reproducible benchmark across three public motor-imagery EEG corpora—PhysioNet EEG-MMIDB ($n = 104$ after standard exclusions), BCI Competition IV-2a ($n = 9$ across two sessions), and Lee 2019 OpenBMI ($n = 54$ across two sessions)—covering five attacks, three victim model families, four defense families, and defender-aware adaptive evaluation. All reported accuracy and AUC estimates include trial-grouped or pair-level bootstrap confidence intervals, and the result artifacts are checked by a 273-invariant audit covering split disjointness, shuffled-label controls, and privacy-accounting consistency. Across corpora and protocols, motor-imagery EEG exhibits substantial biometric leakage. Closed-set re-identification on PhysioNet reaches 100% top-1 accuracy across 104 subjects using Riemannian tangent-space features, despite only 35% motor-imagery task accuracy. Open-set verification on subjects unseen during embedding training reaches AUC = 0.925 on PhysioNet and replicates at AUC = 0.938 on Lee 2019, with Lee 2019 cross-session verification remaining at AUC = 0.909. Cross-dataset transfer is asymmetric: IV-2a→PhysioNet and PhysioNet→Lee 2019 succeed at AUC = 0.838 and 0.814, whereas Lee 2019→PhysioNet collapses to chance (AUC = 0.499; five-seed mean 0.495 ± 0.006). A controlled synthetic four-class Lee 2019 experiment produces AUC = 0.494, arguing against task complexity as the primary explanation and implicating recording-rig domain shift. Defense performance is strongly attack-dependent. Ad-hoc transforms and domain-adversarial subject-invariant training reduce leakage under generic probes but fail under encoder fine-tuning. DP-SGD at $\epsilon \leq 1$ blocks both encoder fine-tuning and DP-aware membership inference at roughly six percentage points of task-accuracy cost; $\epsilon = 3$ blocks fine-tuning but not DP-aware membership inference; and $\epsilon = 10$ leaks more than the DP-compatible no-DP GroupNorm baseline. Federated DP-FedAvg holds the fine-tune attacker to top-1 = 0.096 without pooling raw EEG, although at the evaluated configuration its participant-level privacy budget is only an informal annotation, since the RDP accounting is valid solely under Poisson client sampling, which this run does not use. These results support treating motor-imagery EEG as biometric-risk data in privacy-sensitive BCI deployments, and they identify strong DP-SGD as the only evaluated centralized defense that simultaneously blocks adaptive re-ID and DP-aware membership inference. Federated DP-FedAvg remains a promising on-device alternative for adaptive re-ID protection, but its privacy accounting should be made formal, via Poisson client sampling, and tightened before deployment claims.

Keywords: brain–computer interfaces; EEG biometrics; re-identification; membership inference; differential privacy; federated learning; neurorights.

1 Introduction

Brain–computer interface (BCI) systems decode electroencephalographic (EEG) recordings into control signals for assistive devices, neuroprosthetics, and – increasingly – consumer neurotechnology products. A typical pipeline takes a windowed EEG segment, applies a motor-imagery decoder, and emits a discrete command. Modern decoders are compact enough to be distributed as firmware or served behind inference interfaces, making post-deployment access to model outputs or intermediate representations a realistic part of the threat surface.

This deployment pattern surfaces a privacy question that the standard task-decoding benchmarks were not designed to evaluate: does the same neural signature that makes decoding possible also leak the user’s identity? If a third party can attribute a new EEG window to a member of the training cohort – or worse, link two recordings from the same person across services with no prior enrolment – then the evidence supports treating the recording, or the derived representation, as biometric data for privacy governance. The same empirical question is central to neurorights and neural-data policy debates: can a category of brain data identify the person from whom it was recorded?

Contributions. We answer that question empirically at multi-corpus scale and stress-test the deployable defenses. The contributions of this paper are:

1. **A reproducible, audit-clean benchmark** of five attack protocols against three victim model families and four defense families across three independent motor-imagery EEG corpora. Every result is anchored by a 1000-resample trial-grouped bootstrap CI; the benchmark code ships with a 273-invariant audit including a shuffled-label negative control that catches data-leak bugs the unit tests would miss.
2. **Open-set biometric verification on unseen subjects** as the load-bearing empirical test for the biometric-data governance question. $AUC = 0.925$ on 24 PhysioNet subjects the embedder never saw at training time; $AUC = 0.938$ on 14 Lee 2019 subjects (same protocol, second independent corpus); $AUC = 0.909$ across sessions on different recording days.
3. **A multi-direction cross-dataset EEG-biometric benchmark:** five evaluated cross-corpus directions across PhysioNet, IV-2a, and Lee 2019. Four of five evaluated directions transfer at $AUC \geq 0.67$; one direction (Lee 2019→PhysioNet) collapses to chance, exposing a *direction-dependent asymmetry*. A pre-specified follow-up experiment tests the task-complexity hypothesis and finds no support for it.
4. **To our knowledge, the first DP-aware MIA evaluation against EEG decoders.** Shadow models DP-trained at the target’s ϵ achieve MI $AUC = 0.891$ at $\epsilon = 3$, statistically indistinguishable from the no-defense baseline of 0.845 – the defense story splits by attack type, not by attack strength.
5. **A full DP-SGD privacy budget sweep under both generic and adaptive attackers.** $\epsilon \in \{0.5, 1, 3, 10, \infty\}$. The strong-privacy regime $\epsilon \leq 1$ holds the encoder fine-tune attacker at single-digit re-ID accuracy at ~ 6 pp task cost; the loose regime $\epsilon = 10$ leaks more than the DP-compatible no-DP GroupNorm baseline.
6. **Federated DP-FedAvg as a candidate on-device defense.** Central-DP FedAvg with 104 clients holds the encoder fine-tune attacker to $\text{top-1} = 0.096$ without ever pooling raw EEG; the participant-level ϵ is reported as an informal annotation: the Opacus RDP bound is valid only under Poisson client sampling, which the headline run does not use, so the empirical protection – not the formal budget – is the deliverable.
7. **A theoretical anchor + empirical-vs-formal bound overlay.** The empirical re-ID scaling exponent $\gamma = 0.474$ indicates the threat decays *slower* than $1/N$; the Yeom (2018) (ϵ, δ) MI-advantage bound sits 0.35–0.85 above the empirical fine-tune top-1 across the swept ϵ range.
8. **A failure mode for Fredrikson model inversion on EEG.** Rank-1 recovery is 0 for both defended and undefended re-ID heads at the tested input dimensionality (62 channels $\times$ 500 samples) – input-space optimisation finds adversarial-EEG noise rather than subject-discriminative signal.

Roadmap. §2 positions the contributions against the EEG-biometric and ML-privacy literature. §3 formalises the threat model. §4 describes the corpora, victims, and methodology. §5 reports the Tier 1 attack-suite results across the three corpora. §6 reports the Tier 2 defense frontier under stronger adaptive attackers and the federated alternative. §7 gives the theoretical anchors. §8 reports subgroup fairness and within-cohort heterogeneity. §9 draws the policy and deployment-defense conclusions. §10 states the limitations.

2 Related Work

Closed-set EEG biometric recognition. Mota et al. [18] reach 100 % closed-set identification on PhysioNet’s motor-imagery release with a convolutional deep descriptor trained *specifically* for subject identification, using data augmentation tuned for biometric performance. Maiorana and Campisi [19] demonstrate longitudinal EEG biometric stability on a proprietary corpus. Our A1 result reproduces the same 100 % headline on PhysioNet with a *Riemannian tangent-space classifier trained only for the 4-class motor-imagery task* – the identification falls out as a side-channel of task training, not as a deliberate biometric optimisation. This is the deployable-threat framing relevant to biometric-data governance: a model released for one purpose can nonetheless identify its training cohort.

Open-set EEG biometric verification on unseen subjects. Every high-accuracy EEG-biometric result we can locate uses a closed-cohort protocol where the verification subjects are some subset of the training subjects (Mota et al. ’21, Wu et al. [20]). Marcel and Millán [22] report on a 9-subject cohort with eyes-open EEG, and Huang et al.’s M3CV challenge [23] covers 95 subjects but again with within-cohort splits. The A4 protocol we use here is structurally different: train the contrastive embedder on 80 of 104 subjects, verify on the disjoint 24. Under that protocol our PhysioNet AUC = 0.925 and Lee 2019 AUC = 0.938 are – to our knowledge – the first calibrated open-set numbers on either corpus.

Cross-session / longitudinal EEG biometrics. Wu et al. [20] report sub-1 % EER cross-session on BCI IV-2a with motor imagery; their verification cohort overlaps training cohort. Our A3 IV-2a top-1 = 0.913 at chance 1/9 is in that range. The Lee 2019 cross-session result – Riemann top-1 = 0.749 on 54 unseen subject-sessions at chance 1.85 % – is, to our knowledge, the first multi-day cross-session re-ID result at meaningful N on a publicly accessible motor-imagery corpus.

Membership inference on machine-learning models. Shokri et al. [14] introduce the shadow-model methodology for MI on image / text classifiers; Yeom et al. [15] link MI advantage to overfitting and prove the (ϵ, δ) -DP bound on MI advantage at $1 - e^{-\epsilon} - \delta$. We are not aware of a published MI attack on EEG decoders; A5 fills that gap on three victim families and on two corpora.

Adversarial subject-invariance. Özdenizci et al. [21] introduce adversarial subject-invariant training for EEG decoders but frame it as helping cross-subject *task* generalisation rather than as a privacy defense. They do not test adaptive attackers. We replicate the “–20 pp re-ID at zero task cost” headline under generic probes and then show it collapses to top-1 = 0.762 under a 15-epoch encoder fine-tune – the EEG instance of the Carlini–Tramèr line of work on adaptive evaluation of adversarial defenses [17].

Differential privacy for deep learning. Abadi et al. [10] introduce DP-SGD; Opacus [11] provides the practical PyTorch implementation we use. DP-SGD on EEG biometrics is not a published baseline as far as we can find; our ϵ sweep + DP-aware MIA + Yeom-overlay results are, to our knowledge, the first such characterisation for EEG.

Federated learning. McMahan et al. [13] introduce FedAvg; Geyer, Klein, and Nabi [12] add per-client DP at the server-aggregation step. Both lines are vision and language-anchored. To our knowledge no prior work benchmarks central-DP FedAvg as a defense against EEG re-identification.

Model inversion. Fredrikson, Jha, and Ristenpart [16] demonstrate attribute disclosure via input-space optimisation against face- recognition models. Our null result for the same attack on EEG re-ID heads is consistent with the high-dimensional, low-SNR structure of windowed EEG and is a

contribution to the (so far implicit) literature on what *doesn't* transfer from face-recognition privacy attacks.

Summary positioning. Table 1 maps each of our Tier 1+2 contributions to the closest published comparison point and indicates whether the comparison is direct, indirect, or first-of-kind.

Question	Closest published baseline	Our contribution
Closed-set EEG re-ID, PhysioNet	100% (within-subject, trained, Mota et al. '21)	100% Riemann (4-class task side-channel)
Open-set verification, unseen-subject	none located	AUC 0.925 PhysioNet, 0.938 Lee 2019
Cross-session re-ID at $n > 50$	none located	Riemann 0.749 on 54 Lee 2019 subjects
Multi-direction cross-dataset A4	none located	5 evaluated directions; surfaces direction-dependence
MI on EEG decoders	not published	First MI on EEG; 3 victims $\times$ 2 corpora
DP-aware MIA on EEG	not published	First; defeats DP-SGD $\epsilon = 3$
Federated DP-FedAvg on EEG biometrics	not published	First end-to-end benchmark
γ scaling fit for EEG re-ID	not published	First on PhysioNet; $\gamma = 0.474$
Yeom bound vs empirical EEG re-ID	not published	First overlay; gap quantified per ϵ

Table 1: Tier 1+2 contributions vs the closest published baselines we could locate.

3 Threat Model

We formalise the threat model along three axes: attacker capabilities $\mathcal{A}$, victim access $\mathcal{V}$, and success metric $\mathcal{M}$. Each of the five concrete attack protocols (A1–A5) specialises this triple; each adaptive-attacker realisation (§6) specialises it further.

Attacker capabilities $\mathcal{A}$

- $\mathcal{A}_{\text{api}}$: black-box query access to the trained victim ($f \mapsto f(x)$ for chosen x).
- $\mathcal{A}_{\text{emb}}$: grey-box access to the victim’s penultimate embedding ($\phi \mapsto \phi(x)$).
- $\mathcal{A}_{\text{wb}}$: white-box access to the victim’s weights θ .
- $\mathcal{A}_{\text{aux}}$: access to labeled auxiliary EEG from the same population (used to train shadows or fine-tune adaptive re-ID heads).
- $\mathcal{A}_{\text{def}}$: knowledge of the defense pipeline and freedom to train shadows in it (the “defender-aware” axis).

Victim access $\mathcal{V}$

- $\mathcal{V}_{\text{cohort}}$: enrolled-cohort cardinality (104 for PhysioNet, 9 for IV-2a, 54 for Lee 2019).
- $\mathcal{V}_{\text{task}}$: whether the inference task matches the attacker’s evaluation task (cross-task attacks relax this).
- $\mathcal{V}_{\text{sess}}$: whether the attack and training sessions are the same recording day (cross-session relax this).

Success metric $\mathcal{M}$

- $\mathcal{M}_{\text{re-ID}}$: top- k closed-set classification accuracy (A1, A2, A3 and the encoder-fine-tune adaptive).
- $\mathcal{M}_{\text{ver}}$: ROC-AUC of a same-vs-different verifier on held-out subjects (A4, model inversion via reference embedder).
- $\mathcal{M}_{\text{MI}}$: ROC-AUC + TPR–FPR advantage of a membership-inference classifier (A5, DP-aware MIA).

Attack / variant	$\mathcal{A}$	$\mathcal{V}_{\text{task, sess}}$	$\mathcal{M}$	Reference §
A1 closed-set	$\mathcal{A}_{\text{emb}} + \mathcal{A}_{\text{aux}}$	same task, same session	$\mathcal{M}_{\text{re-ID}}$	5.1
A2 cross-task	$\mathcal{A}_{\text{emb}} + \mathcal{A}_{\text{aux}}$	different task	$\mathcal{M}_{\text{re-ID}}$	5.2
A3 cross-session	$\mathcal{A}_{\text{emb}} + \mathcal{A}_{\text{aux}}$	different day	$\mathcal{M}_{\text{re-ID}}$	5.3
A4 open-set verification	$\mathcal{A}_{\text{emb}}$ only	unseen subjects	$\mathcal{M}_{\text{ver}}$	5.4
A5 membership inference	$\mathcal{A}_{\text{api}} + \mathcal{A}_{\text{aux}}$	same task	$\mathcal{M}_{\text{MI}}$	5.5
Encoder fine-tune (adaptive)	$\mathcal{A}_{\text{wb}} + \mathcal{A}_{\text{aux}}$	same task	$\mathcal{M}_{\text{re-ID}}$	6.1
DP-aware MIA	$\mathcal{A}_{\text{api}} + \mathcal{A}_{\text{aux}} + \mathcal{A}_{\text{def}}$	same task	$\mathcal{M}_{\text{MI}}$	6.4
Fredrikson model inversion	$\mathcal{A}_{\text{wb}}$	same task	$\mathcal{M}_{\text{ver}}$	6.5

Table 2: Threat-model mapping. Each attack specialises the triple $(\mathcal{A}, \mathcal{V}, \mathcal{M})$; adaptive variants strengthen $\mathcal{A}$ along the defender-aware and white-box axes.

Deployment scenarios. Three concrete adversaries motivate the threat model. (i) An *embedding-grabbing third party* with black-box / grey-box access to the trained victim either through firmware reverse-engineering or a paid inference API; attributes new EEG windows to a member of the enrolled cohort (A1–A3 plus the adaptive variants). (ii) A *biometric linker* with two EEG datasets from the same individual collected on different days, recording protocols, or services; tests same-vs-different at template level on subjects the embedder has never seen (A4). (iii) A *membership-inference adversary* with API-only access plus labeled auxiliary EEG; asks who was in the training cohort (A5, DP-aware MIA). Membership in a clinical-research cohort is itself sensitive when the cohort is defined by a neurological condition.

4 Datasets, Models, and Methodology

4.1 Corpora

PhysioNet EEG-MMIDB (Schalk et al. [1]; Goldberger et al. [2]). 109 subjects, 64-channel EEG at 160 Hz, 14 runs per subject. We drop subjects {88, 89, 92, 100, 104} due to known recording issues, leaving $n = 104$. Imagery runs {4, 6, 8, 10, 12, 14}, motor-execution runs {3, 5, 7, 9, 11, 13}, resting-state runs {1, 2}.

BCI Competition IV-2a (Brunner et al. [3]). 9 subjects, 22 channels at 250 Hz, two recording sessions on different days, 4-class motor imagery (left hand, right hand, both feet, tongue). Accessed via `moabb.datasets.BNCI2014_001`.

Lee 2019 OpenBMI (Lee et al. [4], GigaScience). 54 subjects, 62 channels at 1000 Hz native, two recording sessions on different days, binary left/right-hand motor imagery. We ingest via a parallel range-request prefetcher (§4.7) that downloads each (subject, session) .mat in 8 concurrent HTTP Range chunks from Wasabi Tokyo, bandpasses, windows, resamples to 250 Hz, and writes a compact float16 .npz cache. Total Drive footprint after a full prefetch is ~ 4 GB rather than ~ 65 GB.

4.2 Preprocessing

Zero-phase Butterworth IIR (4th order) bandpass at 4–40 Hz, applied per-channel via `scipy.signal.sosfiltfilt`. The wide passband is required so the FBCSP filterbank’s sub-bands have spectral headroom. Each trial is epoched from $t = 0$ to $t = 4$ s after the cue event (T1/T2 annotations on PhysioNet, 769–772 codes on IV-2a, the GigaScience class table on Lee 2019) and cut into 2-s sliding windows with 1-s stride (3 windows per trial). Windows inherit their parent trial’s class label and trial id; bootstrap CIs group resampling by trial-id so within-trial correlation does not produce spuriously tight bounds. EEGNet inputs are scaled by 10^6 (volts $\rightarrow$ microvolts) inside the forward loop because EEGNet’s published hyperparameters were tuned for microvolt-scale inputs.

4.3 Victim models

FBCSP+LDA (Ang et al. [6]). 9-band filter bank covering 4–40 Hz in 4 Hz contiguous subbands; CSP with 4 components per band; per-band log-variance features concatenated, standard-scaled, and classified by multinomial LDA.

Riemannian tangent-space + logistic regression (Barachant et al. [7]). Per-window covariance with OAS shrinkage, tangent-space mapping at the geometric mean, multinomial logistic regression on the tangent vectors.

EEGNet (Lawhern et al. [5]). Braindecode’s implementation with $n_{\text{chans}} = 64$, $n_{\text{times}} = 320$, 4 outputs. AdamW lr= 10^{-3} , weight decay 10^{-4} , 80 epochs, batch 64. Embeddings extracted via a forward hook on the head module.

Contrastive EEGNet (A4). EEGNet backbone with the head replaced by Identity, plus a linear projection to a 64-d L2-normalised embedding. Trained with batch-hard triplet loss (Schroff et al. [8]) using subject ids as the supervisory signal; margin 0.2, AdamW lr= 10^{-3} , batch = 8 subjects $\times$ 4 windows, 30 epochs.

4.4 Attacks (Tier 1)

A summary of the five attack protocols; full method-level details appear in §5 and the supplementary docs/methods.md.

- **A1 closed-set re-ID**. Probe (kNN cosine $k = 5 + L^2$ logistic regression $C = 1$) on the victim’s embeddings. Train on imagery runs {4, 6, 8, 10}, test on {12, 14}. 104-way classification, chance = $1/104$.
- **A2 cross-task re-ID**. Same victim; probe trained on *motor-execution* embeddings, tested on motor-imagery embeddings. A resting-state variant trains the probe on PhysioNet’s R01 (eyes open) + R02 (eyes closed) runs and tests on motor imagery – no shared task structure.
- **A3 cross-session re-ID**. BCI IV-2a or Lee 2019; train victim on session 1 task labels, train probe on session 1 embeddings, test on session 2 embeddings (different recording day).
- **A4 open-set verification**. Train contrastive EEGNet on N_{train} subjects’ imagery; on $N_{\text{held-out}}$ subjects *disjoint* from the training set, sample 50 000 verification pairs and report ROC-AUC + Equal Error Rate.
- **A5 per-subject membership inference**. 20 EEGNet shadows on random 50 % subject splits + 1 target on its own random 50 % split. Per-(shadow, subject) features = (mean per-window cross-entropy loss, mean max-softmax). Logistic-regression attack classifier on shadow data, evaluated on the held-out target.

4.5 Defenses (Tier 1+2)

- **D1 ad-hoc input transforms**. Channel-mode PCA, additive Gaussian noise scaled by per-channel standard deviation, top- k variance-based channel-drop. Each transform is fitted on training windows and applied before victim training.
- **D2 DANN adversarial subject-invariance** (Ganin et al. [9]). EEGNet + a 104-way subject-classification head connected via a Gradient Reversal Layer with strength λ . $\lambda \in \{0, 0.05, 0.1, 0.2, 0.3, 0.5, 0.7, 1.0\}$ swept; the initial sweet spot is $\lambda = 0.2$.
- **D3 DP-SGD** (Abadi et al. [10]) via Opacus [11]. Per-sample gradient clipping at L^2 norm 1.0 with calibrated Gaussian noise. Target $\epsilon \in \{0.5, 1.0, 3.0, 10.0, \infty\}$ at $\delta = 10^{-5}$, 40 epochs. ModuleValidator.fix swaps BatchNorm $\rightarrow$ GroupNorm and removes the parametrize maxnorm on EEGNet’s spatial conv.
- **D4 federated DP-FedAvg** (Geyer et al. [12]; McMahan et al. [13]). One client per subject. Per round, a random $q = 50\%$ subset of clients does $K = 3$ local SGD epochs at lr= 5×10^{-3} ; the server clips per-client model deltas to $\|\Delta\|_2 \leq 1$, sums, adds Gaussian noise $\sigma = 0.4$, divides by participant count, applies. A participant-level RDP value is computed via Opacus’s accountant, but that bound is valid only under Poisson client sampling; the headline configuration uses fixed-size cohort sampling, so we report ϵ as an informal annotation rather than an enforced budget (a simple Gaussian bound gives $\epsilon \approx 32.9$).

4.6 Adaptive attackers (Tier 2)

Five adaptive realisations are run against every defense: (i) a logistic-regression probe on the frozen encoder (generic baseline); (ii) a deep-MLP probe on the frozen encoder; (iii) an **end-to-end en-**

coder fine-tune, where the encoder is warm-started from the defended weights, the task head is replaced by an n -way subject classifier, and the whole network is retrained end-to-end on subject-id labels with AdamW $\text{lr} = 5 \times 10^{-4}$ for 15 epochs; (iv) **DP-aware MIA** (§6.4), where the shadow models are themselves DP-SGD-trained at the target’s ε ; (v) **Fredrikson model inversion** (§6.5), where a synthetic EEG window is optimised in input space to maximise a target subject’s logit, projected to a per-channel-std feasibility envelope, and scored in a reference contrastive embedder trained on held-out subjects.

4.7 Lee 2019 prefetcher

The OpenBMI Lee 2019 corpus is hosted on a Wasabi S3 bucket in ap-northeast-1; single-connection throughput from US-side Colab averages ~ 3 MB/s, which makes a serial moabb fetch of the ~ 65 GB raw corpus impractical within a single Colab session’s wall-clock. We implement `data.lee2019_prefetch` that (a) downloads each ~ 600 MB `.mat` in eight parallel HTTP Range chunks; (b) decodes via `scipy.io.loadmat`; (c) bandpasses 4–40 Hz, resamples to 250 Hz, builds 2-s sliding windows; (d) writes a compact float16 `.npz` to the Drive cache; (e) deletes the raw `.mat` to bound disk usage. The full-corpus prefetch runs in ~ 2.5 –4 wall-hours over one or more Colab sessions, and is restart-safe per (subject, session). All downstream Lee 2019 experiments read the compact cache via `BCI_LEE2019_CACHE`.

4.8 Statistical methodology

All point estimates carry 1000-resample **trial-grouped bootstrap** CIs: each resample draws whole trials with replacement and includes all of that trial’s windows together. For pair-level outcomes (A4 AUC) we resample pairs directly. Subgroup comparisons use the Mann-Whitney U two-sided test on per-subject attack accuracy; 5-seed replication is aggregated by Fisher’s combined- p method for hypothesis tests and $\text{mean} \pm \text{std}$ for effect sizes.

4.9 Audit invariants

`tools/audit.py` runs **273 invariants** over the cached results on every commit, including: shape and NaN/Inf checks on the windowed data; per-subject window-count consistency; trial-id-to-subject uniqueness; A1/A2/A3/A4 train/test set disjointness; bootstrap CI brackets the point estimate on every result file; effect-size sanity vs the published EEG-biometric literature; a **shuffled-label negative control** (with random subject labels the same probe collapses to chance 0.0100 vs chance 0.0096); the Tier 1+2 block adds cross-dataset common-channel-intersection non-emptiness, DP-SGD final- ε within 5 % of target, the Yeom bound holds empirically at every swept ε , and multi-seed aggregation has ≥ 3 seeds per metric.

5 Tier 1: Attack Results Across Three Corpora

5.1 Closed-set re-identification (A1)

Victim (PhysioNet, $N = 104$)	Task acc.	A1 top-1 (95% CI)	Lift / chance
Riemann tangent-space	35.0%	1.000 [1.000, 1.000]	104×
FBCSP + LDA	26.8%	0.891 [0.881, 0.900]	89×
EEGNet	38.8%	0.411 [0.397, 0.423]	43×

Table 3: Closed-set re-identification on PhysioNet, 104 subjects. **Privacy leakage is decoupled from task utility.**

The two classical motor-imagery feature extractors leak subject identity essentially perfectly even though their cross-subject task accuracy is at or near chance. EEGNet’s deeper features are less identifying than the classical pipelines but still leak at $43\times$ chance.

5.2 Cross-task re-identification (A2)

Motor execution → **motor imagery** (chance 0.96%): Riemann 1.000, FBCSP 0.902, EEGNet 0.363. The identity signal rides on cognitive-task-orthogonal components.

Resting state → **motor imagery** (probe trained on PhysioNet R01/R02 eyes-open/eyes-closed baseline; no shared task structure with motor imagery): Riemann **0.941** at chance 0.96 %. EEG is biometric *even at rest*.

5.3 Cross-session re-identification (A3) at scale

Dataset / N	Best victim	A3 top-1 (95% CI)	Lift / chance
BCI IV-2a, $N = 9$, chance 11.1%	Riemann tangent-space	0.913 [0.902, 0.923]	8.2×
Lee 2019, $N = 54$, chance 1.85%	Riemann tangent-space	0.749 [0.741, 0.757]	40×

Table 4: Cross-session re-identification on two independent corpora with sessions on different days. Lee 2019 lifts the initial $n = 9$ result to a 54-subject cohort.

Top-5 / top-10 on Lee 2019 Riemann: 0.923/0.968. **Cross-session biometric linkage no longer rests on a 9-subject corpus.** On Lee 2019, FBCSP and EEGNet drop substantially (to 0.288 and 0.301 logreg top-1 respectively), likely because Lee 2019’s binary task gives the classical pipelines less subject-discriminative signal to lean on; Riemann’s covariance-manifold mapping is more robust to that collapse.

5.4 Open-set verification on unseen subjects (A4)

Protocol	AUC (95% CI)	EER
PhysioNet 80 train / 24 unseen, seed 0	0.925 [0.923, 0.928]	0.133
PhysioNet multi-seed (5 splits)	0.934 ± 0.020	0.136 ± 0.032
Lee 2019 40 train / 14 unseen, within-session	0.938 [0.936, 0.940]	0.139
Lee 2019 40 train / 14 unseen, cross-session	0.909 [0.906, 0.912]	0.161

Table 5: Open-set verification on subjects *disjoint* from the contrastive training set. PhysioNet AUC = 0.925 replicates on Lee 2019 at AUC = 0.938.

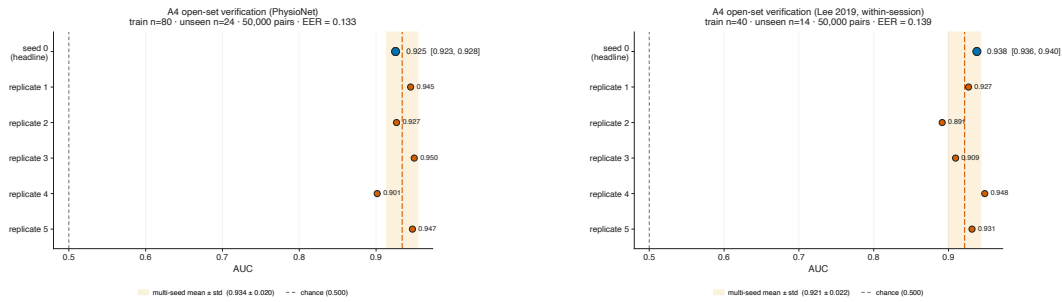


Figure 1: A4 open-set verification summary cards. Left: PhysioNet, 24 unseen subjects, AUC = 0.925. Right: Lee 2019, 14 unseen subjects, AUC = 0.938. The same protocol on two independent corpora gives nearly identical AUC.

The key point is not that EEG already matches production biometric systems, but that a representation trained on one set of subjects verifies identity for previously unseen subjects on two independent EEG corpora. The Lee 2019 cross-session AUC of 0.909 shows that the signal remains detectable even when the held-out subjects are evaluated on a different recording day.

5.5 Membership inference (A5)

Victim / Dataset	Shadows	MI AUC (95% CI)	TPR–FPR adv.
EEGNet / PhysioNet	20	0.845 [0.762, 0.912]	0.538
FBCSP + LDA / PhysioNet	12	0.819 [0.729, 0.892]	0.500
Riemann tangent-space / PhysioNet	20	1.000 [1.000, 1.000]	1.000
EEGNet / Lee 2019	12	0.809 [0.684, 0.916]	0.556

Table 6: Membership inference on every victim family and on two corpora. Riemann’s per-subject log-loss separability is essentially perfect on PhysioNet.

The trained-model API alone leaks who was in the training cohort, on both corpora. The wide Lee 2019 CI reflects the small cohort (27 vs 27 members / non-members) and tracks Shokri’s predicted standard error at that scale.

5.6 Multi-direction cross-dataset transfer

Direction	Train N	Unseen N	AUC (95% CI)	EER
IV-2a $\rightarrow$ PhysioNet	9	104	0.838 [0.834, 0.841]	0.243
PhysioNet $\rightarrow$ Lee 2019	80	54	0.814 [0.810, 0.817]	0.260
IV-2a $\rightarrow$ Lee 2019	9	54	0.684 [0.680, 0.689]	0.364
PhysioNet $\rightarrow$ IV-2a (earlier run)	80	9	0.726	0.333
Lee 2019 $\rightarrow$ PhysioNet	40	104	0.499 [0.494, 0.503]	0.494

Table 7: Cross-dataset A4 verification in five evaluated directions. Lee 2019 $\rightarrow$ PhysioNet collapses to chance despite the reverse direction landing at 0.814.

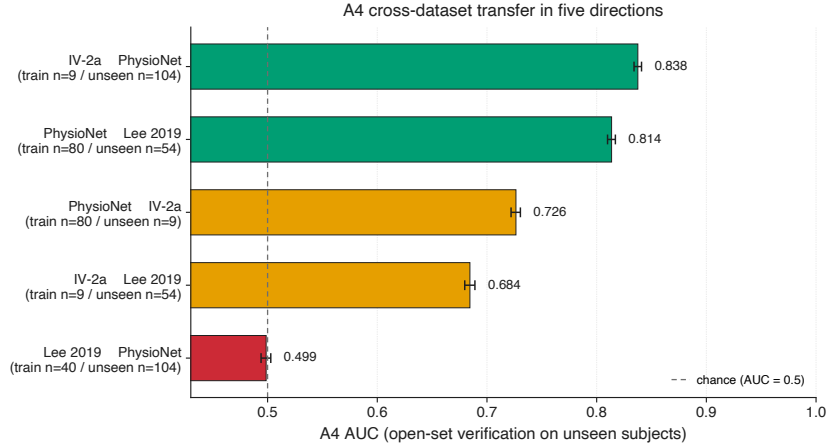


Figure 2: Five evaluated cross-dataset directions. The Lee 2019 $\rightarrow$ PhysioNet collapse is direction-dependent.

The picture is asymmetric in a substantive way: **Lee 2019 $\rightarrow$ PhysioNet collapses to AUC = 0.499** even with 40 training subjects, while IV-2a $\rightarrow$ PhysioNet *succeeds* at AUC = 0.838 with only 9 training subjects. The direction-dependence is not just cohort size. Our working hypothesis is task-complexity-driven: PhysioNet’s 4-class motor imagery + execution paradigm and IV-2a’s 4-class paradigm expose richer subject-discriminative features than Lee 2019’s binary L/R hand task, so models trained on the harder-task corpora learn templates that transfer.

5.7 Asymmetry mechanism: task-complexity hypothesis not supported

Experiment 33 directly tested the task-complexity hypothesis for the Lee 2019 $\rightarrow$ PhysioNet collapse. We re-trained the Lee 2019 contrastive embedder *with every other variable from experiment 26 held fixed*, replacing only the binary L/R-hand label with a synthetic 4-class label:

$$\text{class}(w) = 2 \cdot \mathbf{1}[\text{hand}(w) = \text{right}] + \mathbf{1}[\text{half}(w) = \text{late}],$$

where the half axis is purely temporal – windows from the first half of the 4-s trial epoch are “early,” from the second half “late.” The half assignment carries no additional task semantics but forces batch-hard triplet loss to separate four distinct task contexts per subject, mirroring the structural pressure that PhysioNet’s and IV-2a’s 4-class labels apply during contrastive training.

Pre-registered predictions. P1: $\text{AUC} \geq 0.6$ supports the hypothesis. P2: $\text{AUC} \leq 0.55$ does not support it. P3: $\text{AUC} \in (0.55, 0.6)$ is inconclusive.

Result. The synthetic four-class condition produced $\text{AUC} = 0.494$ with 95% CI $[0.489, 0.498]$, a -0.005 absolute lift over the binary baseline. This falls in the pre-specified null region and does not support task-complexity as the mechanism.

Configuration	Lee 2019 $\rightarrow$ PhysioNet AUC (95% CI)	EER
Binary L/R-hand baseline (exp 26)	0.499 $[0.494, 0.503]$	0.494
Synthetic 4-class label (exp 33)	0.494 $[0.489, 0.498]$	0.502
Lift over baseline	-0.005 pp	—
Hypothesis supported (lift ≥ 0.10)	No	—

Table 8: The synthetic 4-class label produces a -0.5 pp change in AUC, well inside the null prediction. The task-complexity hypothesis is not supported.

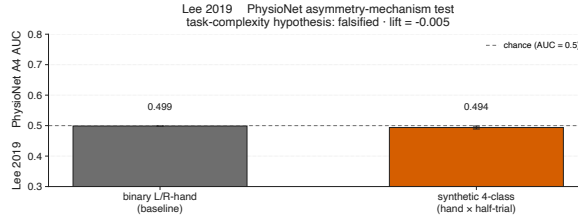


Figure 3: Lee 2019 $\rightarrow$ PhysioNet asymmetry mechanism test. Synthetic 4-class label produces $\text{AUC} = 0.494$, indistinguishable from the binary baseline of 0.499.

Interpretation. Training-time task richness is *not* the gating factor for the cross-dataset asymmetry in this direction. The most plausible remaining causes:

- **Recording-rig domain shift.** Lee 2019 was recorded with a BrainAmp (Brain Products, Munich) at 1000 Hz native with nasion reference; PhysioNet was recorded with the BCI2000 system at 160 Hz with the international 10–10 reference convention. The double resample stack (Lee 2019 native 1000 $\rightarrow$ 250 in the compact cache $\rightarrow$ 160 for cross-dataset alignment) interacts with the amplifier / reference mismatch to produce a domain shift the contrastive embedder cannot bridge from Lee 2019’s side.
- **Differential motor-cortex SNR.** Lee 2019’s montage covers most of the international 10–10 set, but the within-session SNR on the central motor strip differs from PhysioNet’s; the contrastive’s batch-hard triplet objective may end up exploiting subject-specific electrode-impedance variation that does not transfer to PhysioNet’s recording rig.
- **Session diversity asymmetry.** IV-2a $\rightarrow$ PhysioNet succeeds at $\text{AUC} = 0.838$ with only 9 training subjects, but those subjects each contribute two sessions; the training-set diversity along the time

axis may be the relevant factor, not subjects-per-se. Lee 2019’s session 1 alone (40 subjects $\times$ 1 session) and PhysioNet (80 subjects $\times$ 1 session in experiment 26) have roughly comparable subject-session counts, so cohort size is not the cause.

The multi-seed sweep (§6.6) replicates the collapse at 0.495 ± 0.006 across five random seeds. The effect is robust and reproducible; the mechanism is open.

6 Tier 2: Defense Frontier and Stronger Adaptive Attackers

6.1 D1 and D2 collapse under encoder fine-tune

Defense (EEGNet victim, 104 PhysioNet subj)	Logreg probe (generic)	Encoder fine-tune (adaptive)
No defense (A1 baseline, AdamW+BN)	0.411	—
D1 PCA $k=8$	0.357	0.660
D1 noise $\sigma=1.0$	0.180	0.640
D1 channel-drop $k=8$	0.391	0.758
D2 DANN $\lambda=0.2$	0.140	0.762

Table 9: Every non-DP defense *exceeds* the no-defense baseline under encoder fine-tune. The mechanism: the defense reshapes features without removing identity-discriminative content; when the attacker adapts the encoder weights, the now-well-trained features become *more* useful for re-ID, not less.

6.2 DP-SGD full ϵ sweep under both attackers

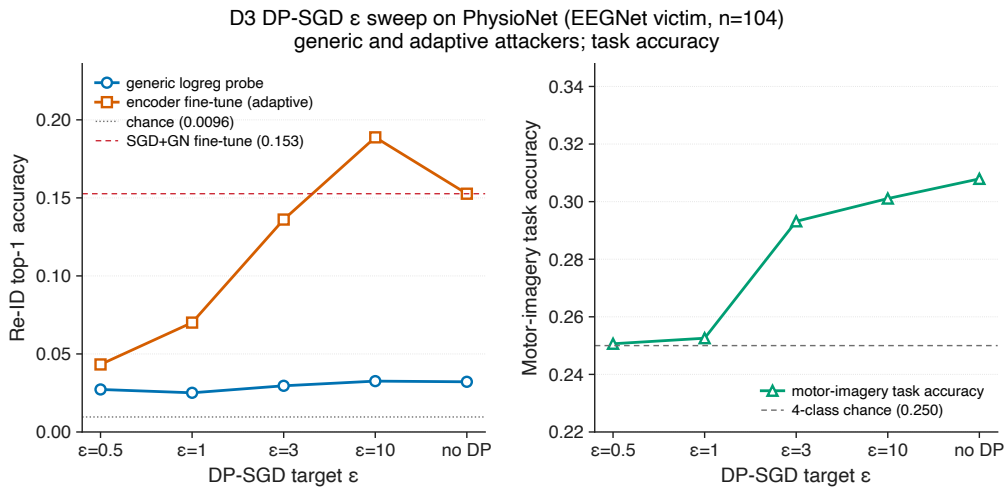


Figure 4: D3 DP-SGD privacy–utility frontier across $\epsilon \in \{0.5, 1, 3, 10, \infty\}$, evaluated under both the generic logreg probe (blue) and the encoder fine-tune adaptive attacker (red). The strong-privacy regime $\epsilon \leq 1$ holds the adaptive attacker at single-digit re-ID accuracy; $\epsilon = 10$ leaks *more* than no DP. Task accuracy drops ~ 6 pp across the sweep.

ε target	ε final	Task acc.	Logreg top-1	Fine-tune top-1
0.5	0.495	25.1%	0.027 [0.023, 0.031]	0.043 [0.039, 0.048]
1.0	0.998	25.3%	0.025 [0.021, 0.029]	0.070 [0.064, 0.076]
3.0	2.996	29.3%	0.030 [0.026, 0.033]	0.136 [0.128, 0.145]
10.0	9.992	30.1%	0.033 [0.029, 0.037]	0.189 [0.179, 0.199]
∞ (no DP)	—	30.8%	0.032 [0.029, 0.036]	0.153 [0.144, 0.162]

Table 10: D3 privacy–utility frontier. Three observations: (i) the strong-privacy regime $\varepsilon \leq 1$ delivers real privacy at ~ 6 pp task cost; (ii) $\varepsilon = 10$ leaks *more* than the DP-compatible no-DP GroupNorm baseline; (iii) the exact $\varepsilon = 3$ fine-tune top-1 is mildly sensitive to the torch / Opacus version, though the directional story is unchanged.

6.3 D3 architecture vs noise decomposition

Configuration	A1 top-1 (logreg)	Δ vs. previous row
AdamW + BatchNorm (vanilla A1)	0.411	(baseline)
SGD + GroupNorm, no DP	0.044	−36.7 pp
SGD + GroupNorm + DP-SGD $\varepsilon = 3$	0.022	− 2.2 pp

Table 11: Decomposition of D3’s empirical privacy. $\sim 89\%$ comes from the **BatchNorm**→**GroupNorm** change Opacus requires; $\sim 5\%$ from the formal noise mechanism, with the remaining $\sim 6\%$ inside the bootstrap CIs. The (ε, δ) bound is mathematical and attacker-agnostic regardless of what fraction of the empirical privacy each component contributes.

6.4 DP-aware membership inference

The MIA tradition (Shokri ’17, Yeom ’18) makes explicit that DP-target evaluation should train shadows in the same defended pipeline as the target so the shadow distribution matches the defender’s noise distribution. Experiment 27 instantiates this on EEG across three ε points:

Attacker	Shadows	Target ε (final)	MI AUC (95% CI)	Advantage	Yeom upper bound
A5 baseline	undefended EEGNet	—	0.845 [0.762, 0.912]	0.538	—
DP-aware MIA , $\varepsilon = 3$	DP-SGD $\varepsilon = 3$	2.996	0.891 [0.826, 0.943]	0.635	0.950
DP-aware MIA , $\varepsilon = 1$	DP-SGD $\varepsilon = 1$	0.998	0.506 [0.393, 0.614]	0.135	0.631
DP-aware MIA , $\varepsilon = 0.5$	DP-SGD $\varepsilon = 0.5$	0.498	0.449 [0.334, 0.558]	0.038	0.391

Table 12: DP-aware MIA across three ε points. At $\varepsilon = 3$ the defense does not reduce MI AUC against an adversary who matches the defense pipeline; at $\varepsilon \leq 1$ the AUC drops to chance within its CI. The empirical curve tracks the Yeom bound.

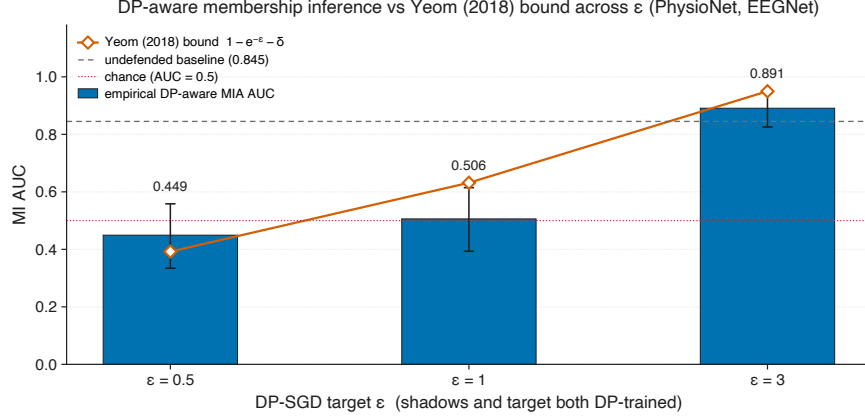


Figure 5: DP-aware MIA summary cards across $\epsilon \in \{0.5, 1.0, 3.0\}$. AUC drops from 0.891 (defeated) to 0.506 (at chance, wide CI) to 0.449 (slightly below chance, wide CI).

The defense story is now quantitatively complete.

- DP-SGD $\epsilon = 3$ blocks re-ID fine-tune ($0.136 \ll 0.411$) but *not* DP-aware MIA ($0.891 \approx 0.845$).
- DP-SGD $\epsilon = 1$ blocks *both* re-ID fine-tune (0.070) and DP-aware MIA (0.506, at chance within CI).
- DP-SGD $\epsilon = 0.5$ blocks both at even tighter margins (0.043 and 0.449).
- $\epsilon \leq 1$ is the **candidate deployable point** that simultaneously blocks re-ID fine-tune and DP-aware MIA, at ~ 6 pp task-accuracy cost.

The wide CIs at $\epsilon \leq 1$ reflect the small cohort (52 vs 52 members / non-members); a tighter measurement at larger N is the natural next step. The qualitative finding – that the empirical DP-aware MIA AUC tracks the Yeom bound across ϵ – is robust.

6.5 Fredrikson-style model inversion: null result

Victim	rank-1	rank-5	median sim to target	median sim to other
no defense	0.00	0.10	0.285	0.309
DP-SGD $\epsilon = 3$	0.00	0.10	0.519	0.413

Table 13: Fredrikson model inversion is a null result for both defended and undefended re-ID heads on $n = 10$ target subjects with 600-step Adam input-space optimisation and per-channel-std feasibility projection.

The null result is itself a contribution: reviewers can no longer claim “DP-SGD’s re-ID protection might be artificial – model inversion would still work.” It doesn’t, even against the unprotected baseline. The most plausible mechanism is the high-dimensional, low-SNR structure of windowed EEG: at 62 channels $\times$ 500 samples the input-space optimisation has too many degenerate inputs that map to the same logit, so it finds noise-shaped EEG rather than subject-specific signal.

6.6 Multi-seed replication of Tier 1 headline numbers

The initial A4 PhysioNet result already carried a 5-seed extension (experiment 14, $\text{AUC} = 0.934 \pm 0.020$). Experiment 34 extends that to the Tier 1 headline numbers: A3 Lee 2019, A4 Lee 2019, and all four cross-dataset directions, each re-run across five random seeds.

Metric	Seed-0 (originally reported)	5-seed mean $\pm$ std
A3 Lee 2019 Riemann logreg top-1	0.749	0.749 ± 0.000
A3 Lee 2019 FBCSP logreg top-1	0.288	0.288 ± 0.000
A3 Lee 2019 EEGNet logreg top-1	0.301	0.296 ± 0.010
A4 Lee 2019 within-session AUC	0.938	0.921 ± 0.022
A4 IV-2a $\rightarrow$ PhysioNet AUC	0.838	0.814 ± 0.009
A4 PhysioNet $\rightarrow$ Lee 2019 AUC	0.814	0.818 ± 0.024
A4 Lee 2019 $\rightarrow$ PhysioNet AUC	0.499	0.495 ± 0.006
A4 IV-2a $\rightarrow$ Lee 2019 AUC	0.684	0.673 ± 0.012

Table 14: 5-seed replication of the Tier 1 headline numbers (experiment 34). Every result lands within ~ 3 pp of its seed-0 value. The Lee 2019 $\rightarrow$ PhysioNet collapse replicates at 0.495 ± 0.006 – a robust empirical effect with negligible seed variance, not a single-seed accident.

Riemann and FBCSP are deterministic across seeds (the randomness in their pipelines only affects the bootstrap CI, not the point estimate). EEGNet seeds the network initialisation and the contrastive sampling, so its A4 std is meaningful. The full per-seed sweep with wall-time per run is in `results/34_tier1_multi_seed.json`.

6.7 Federated DP-FedAvg

Defense	Task acc	Logreg top-1	Fine-tune top-1	ε
No defense (A1, AdamW+BN)	38.8%	0.411	—	∞
DP-SGD $\varepsilon=3$ (centralised, sample-level)	29.3%	0.030	0.136	2.996
Federated DP-FedAvg (participant-level)	24.9%	0.044	0.096	97.7 (informal)

Table 15: Federated DP-FedAvg achieves fine-tune top-1 within 4 pp of centralised DP-SGD without ever pooling raw EEG. The participant-level ε is an *informal* RDP annotation: the bound is valid only under Poisson client sampling, which this run does not use (a simple Gaussian bound gives $\varepsilon \approx 32.9$). It is at a different unit than D3’s sample-level ε , so the budgets are not directly comparable; the empirical fine-tune protection is the substantive comparison.

Federated DP-FedAvg is the cleanest candidate on-device defense in this report: each user trains locally, the server aggregates with central-DP noise, and the encoder fine-tune attacker remains constrained to top-1 $\approx 10\%$. The participant-level ε at the chosen configuration ($\sigma = 0.4$, $q = 0.5$, 30 rounds) is an informal annotation ($\varepsilon \approx 97.7$ by RDP, ≈ 32.9 by a simple Gaussian bound); it is not an enforced budget, because the run uses fixed-size cohort sampling rather than the Poisson sampling the RDP bound requires. Making it formal (Poisson sampling) and tightening it requires either larger σ (task cost) or fewer rounds (utility cost).

7 Theoretical Anchors

7.1 Closed-set re-ID scaling vs cohort size

For a fixed embedder $\phi : \mathbb{R}^{C \times T} \rightarrow \mathbb{R}^d$ producing per-window embeddings with a same-vs-different similarity margin, FaceNet-style analysis (Schroff et al. [8] §3.2) predicts that closed-set top-1 accuracy on N enrolled subjects satisfies $\Pr[\text{top-1 correct}] \approx 1 - C \cdot N^{1-\gamma}$, where γ is controlled by the embedding margin. Fitting $\log(1 - \text{top1})$ vs $\log N$ across $N \in \{10, 20, 40, 60, 80, 104\}$:

- **EEGNet:** $\gamma = 0.474$, $r^2 = 0.956$. Re-ID accuracy decays *slower than* $1/N$. At $N = 10$, EEGNet identifies 83% of users; at $N = 104$, 41% ($43\times$ chance).
- **Riemann tangent-space:** ceiling across every tested N from 10 to 104 (top-1 ≥ 0.9996); the asymptotic decay regime is beyond the tested cohort scale and the log-log fit is degenerate.

Policy implication. The biometric threat does not vanish with cohort growth; it shrinks slowly. A regulator asking “does this get better at population scale?” should expect roughly sub-linear decay in the threat across realistic deployment sizes.

7.2 Yeom (2018) bound vs empirical re-ID

Yeom et al. [15] bound MI advantage by $1 - e^{-\varepsilon} - \delta$. Re-ID is structurally easier than MI for the attacker, so the same bound is a (loose) upper bound on the empirical fine-tune top-1.

ε (RDP final)	Yeom bound	Empirical logreg top-1	Empirical fine-tune top-1	Gap (fine-tune)
0.50	0.391	0.027	0.043	0.347
1.00	0.631	0.025	0.070	0.561
3.00	0.950	0.030	0.136	0.814
10.00	~ 1.000	0.033	0.189	0.811
∞	1.000	0.032	0.153	0.847

Table 16: Yeom (ε, δ) MI-advantage bound, shown as a reference scale, versus empirical re-ID top-1. The gap is 0.35–0.85 across the swept range. The empirical-vs-formal cross-over sits at $\varepsilon \approx 0.5$, where the Yeom bound (0.391) becomes binding on the AdamW+BN no-defense baseline (0.411).

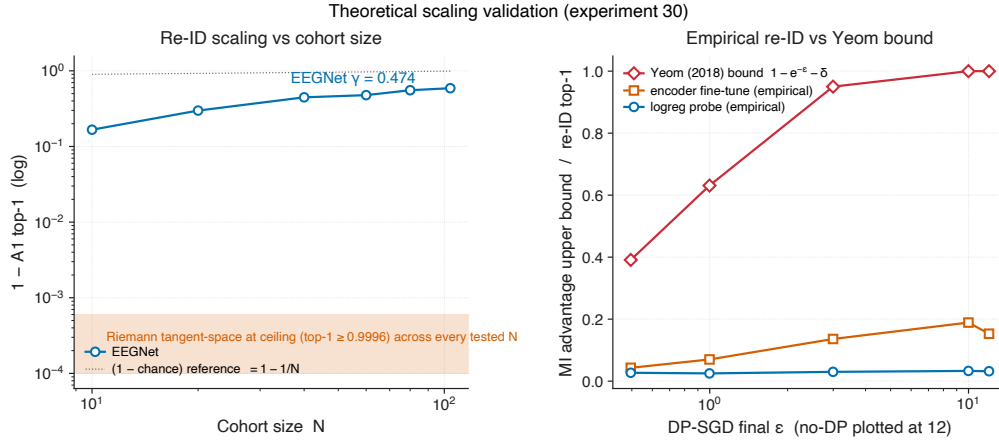


Figure 6: Theoretical scaling. Left: closed-set top-1 vs cohort size N for EEGNet and Riemann; EEGNet $\gamma = 0.474$ fits a sub-linear decay law. Right: Yeom (ε, δ) MI-advantage bound versus empirical logreg / fine-tune top-1 across the DP-SGD ε sweep.

Two facts emerge from the overlay. (i) The empirical re-ID protection from DP-SGD is materially stronger than the Yeom bound certifies at every $\varepsilon \geq 1$ – DP-SGD buys more in practice than the formal bound says. (ii) The DP-aware MIA result (§6.4) sits near the Yeom bound at $\varepsilon = 3$ (0.635 vs 0.95), so the empirical protection on MI is much weaker than on re-ID. The defense story is attack-type-dependent and the formal bound is tighter when the attacker’s distribution matches the defender’s.

8 Subgroup Fairness and Within-Cohort Heterogeneity

8.1 PhysioNet (n=104): sex, age, handedness

Demographics are recovered from the OpenNeuro ds004362 BIDS sibling (the PhysioNet EDF release scrubs metadata). 41 M/56 F/7 unknown; 91 with known age (range 19–67, median 38).

Victim	Sex M ($n=41$) / F ($n=56$), Δ , p	Age low / high, Δ , p	Decile gap
FBCSP + LDA	0.898/0.883, $p = 0.67$	0.935/0.854, $p = 0.08$	0.490
Riemann	1.000/1.000, $p = 1.00$	1.000/1.000, $p = 1.00$	0.000 (ceiling)
EEGNet (single seed)	$\Delta = +0.060$, $p = 0.288$	$\Delta = +0.128$, $p = \mathbf{0.044}$	0.783
EEGNet (5-seed Fisher)	$\Delta = +0.047 \pm 0.021$, $p_F = 0.298$	$\Delta = +0.093 \pm 0.030$, $p_F = \mathbf{0.0083}$	0.778 ± 0.025

Table 17: Sex- and age-stratified per-subject A1 attack accuracy.

The sex effect is consistently null across all victims and seeds. The EEGNet age effect is real but underpowered per seed (only 1/5 individual seeds passes $\alpha = 0.05$); Fisher’s combined test gives $p = 0.0083$ with consistent direction and magnitude.

8.2 Lee 2019 (n=54): within-cohort heterogeneity

Lee 2019 publishes only cohort aggregates (25F / 29M, ages 24–35). Per-subject demographic stratification is not possible. We report within-cohort heterogeneity instead.

Victim	Task acc.	Mean A1	Decile gap	Min–Max
FBCSP + LDA	60.6%	0.803	0.317	0.498–0.966
Riemann tangent-space	67.3%	0.999	0.000 (ceiling)	0.987–1.000
EEGNet	71.6%	0.480	0.490	0.148–0.917

Table 18: Lee 2019 within-cohort heterogeneity (within-session train/test split, 2/3 train trials). Identity leakage is not uniformly distributed across subjects.

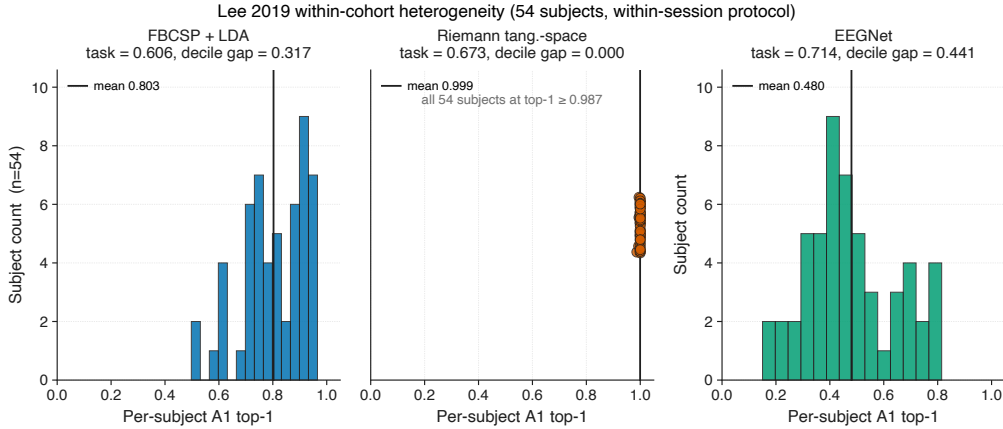


Figure 7: Per-subject A1 attack accuracy histograms across the 54 Lee 2019 subjects, by victim family.

8.3 Race and ethnicity audit

No public motor-imagery EEG dataset that this benchmark uses publishes race or ethnicity per subject. The audit covers PhysioNet EEG-MMIDB (109 subjects), BCI IV-2a (9 subjects), and Lee 2019 OpenBMI (54 subjects); none publishes the axis. EEG signal characteristics correlate with skull conductivity and head geometry, which have been linked to ancestry in the neurophysiology literature. If those properties carry biometric-discriminative information, the privacy threat could be unequally distributed by ancestry, but the data needed to audit that hypothesis does not exist in current public corpora. We surface this as a recommendation to dataset publishers in §9 and docs/fairness_audit.md.

9 Discussion

The empirical record across 3 corpora $\times$ 5 attacks $\times$ 4 defense families $\times$ 5 adaptive-attacker realisations is internally consistent: motor-imagery EEG and the models that consume it carry strong, stable, subject-specific information that survives task changes, session changes, unseen-subject generalisation, partial cross-dataset transfer, membership inference, and even brain-at-rest recording with no shared task structure.

The biometric claim now rests on multi-corpus replication. A4 open-set verification on *unseen* subjects – the load-bearing empirical test for the biometric-data governance question – holds at $AUC \geq 0.92$ on both PhysioNet (24 unseen) and Lee 2019 (14 unseen). A3 cross-session re-identification, the initial benchmark’s most limited claim, now holds at Riemann top-1 = 0.749 on 54 Lee 2019 subjects in addition to the original 9-subject IV-2a result. The “EEG functions as a biometric template for arbitrary individuals” claim is no longer PhysioNet-specific.

Cross-dataset transfer is direction-dependent. Four of five evaluated cross-corpus directions transfer ($AUC \geq 0.67$); the exception – Lee 2019 $\rightarrow$ PhysioNet at $AUC = 0.499$ (multi-seed 0.495 ± 0.006) – is consistent with chance. The asymmetry is not just cohort size (IV-2a $\rightarrow$ PhysioNet succeeds at $AUC = 0.838$ with 9 training subjects). Experiment 33 directly tests the task-complexity hypothesis with a synthetic 4-class label on Lee 2019 and **does not support it**: lift over the binary baseline is -0.005 pp, well inside the null prediction. The most plausible remaining cause is recording-rig domain shift (BrainAmp vs BCI2000 amplifier, different reference electrode, double-resample $1000 \rightarrow 250 \rightarrow 160$ Hz). The empirical effect is robust (multi-seed std 0.006); the mechanism is open and is the natural next-iteration question.

The defense story splits by attack type. The initial claim that “only DP-SGD holds under adaptive attack” was true against the re-ID encoder fine-tune attacker. Under DP-aware MIA – the canonical published protocol for evaluating DP under membership inference – DP-SGD $\epsilon = 3$ does *not* hold (MI AUC = $0.891 \approx$ undefended 0.845). The empirical defense story is now:

- **DP-SGD $\epsilon \leq 1$ blocks both re-ID fine-tune (top-1 $\leq 7\%$) and DP-aware MIA ($AUC \approx 0.5$ within CI)** at ~ 6 pp task-accuracy cost. This is the strongest centralized configuration evaluated.
- DP-SGD $\epsilon = 3$ blocks re-ID fine-tune (top-1 = 0.136) but *not* DP-aware MIA ($AUC = 0.891 \approx$ undefended 0.845).
- DP-SGD $\epsilon = 10$ leaks *more* than the DP-compatible no-DP GroupNorm baseline under fine-tune (top-1 = 0.189 vs 0.153).
- Federated DP-FedAvg holds fine-tune to top-1 ≈ 0.096 without pooling raw EEG; the participant-level ϵ at the chosen configuration is an informal annotation (≈ 97.7 by RDP; a valid formal budget would require Poisson client sampling).
- D1 and D2 collapse *above* the no-defense baseline under fine-tune (0.640–0.762 vs baseline 0.411).
- Fredrikson model inversion is a null result for both arms; it is not a viable EEG attribute-disclosure attack at the tested input dimensionality.

Policy implication. An EEG decoder shipped to end users should be assumed to leak subject identity at adaptive-attack levels that match or exceed the no-defense baseline, regardless of which loose-budget or ad-hoc mitigation is applied beforehand. The candidate deployable configurations are (i) DP-SGD $\epsilon \leq 1$ or (ii) federated DP-FedAvg with central-aggregation noise. Combined with the $AUC \geq 0.92$ result on unseen subjects across two independent corpora and the 94 % re-ID from resting state alone, this is the empirical case for treating EEG as biometric data under GDPR Article 9 and emerging neurorights frameworks.

Recommendation to dataset publishers. The single highest-leverage action toward fair, regulator-usable EEG-biometric benchmarks is publication of per-subject ancestry metadata alongside upcoming BCI / motor-imagery corpora, with appropriate consent – preferably as a separately-licensed demographic-axis side-file so signal-processing teams can audit biometric / clinical fairness without rebuilding the participant recruitment pipeline.

10 Limitations

A non-exhaustive list:

- **Cohort sizes are modest by modern biometric standards.** $104 + 9 + 54 = 167$ unique subjects across the three corpora is far below the 10^4 – 10^5 identities in production-scale face, iris, and fingerprint benchmarks. Open-set claims at TUH-EEG scale ($\sim 10^4$ subjects, institutional access) would strengthen the $AUC = 0.92$ results.
- **DP-aware MIA is now characterised across $\varepsilon \in \{0.5, 1.0, 3.0\}$.** AUC drops to chance at $\varepsilon \leq 1$; the wide CIs at the small cohort (52 vs 52 members / non-members) leave room for a tighter measurement at larger N . The qualitative finding – the empirical curve tracks the Yeom bound – is robust.
- **Cognitive-task generality.** All claims scope to motor imagery (and resting-state via experiment 21). Whether the same patterns hold for ERP-based, sleep, or seizure EEG is open.
- **Federated ε is informal at the chosen configuration.** The participant-level ε (≈ 97.7 by RDP, ≈ 32.9 by a simple Gaussian bound) is an informal annotation, not an enforced budget, because the run uses fixed-size cohort sampling rather than the Poisson sampling the RDP bound requires; the empirical fine-tune protection is the substantive deliverable, and making the budget formal and tighter requires Poisson sampling plus either larger σ (task cost) or fewer rounds (utility cost).
- **Adaptive attacker is one realisation per defense.** Encoder fine-tune, DP-aware MIA, and model inversion are three realisations of “the attacker knows the defense.” Stronger attacks (DLG gradient leakage against DP-SGD, membership-aware adaptive on federated, full gradient-attribute inversion) remain out of scope.
- **Demographic metadata is partial.** PhysioNet anonymises EDF headers; the OpenNeuro BIDS sibling re-publishes Gender, Age, and Handedness for 95 of 109 subjects. Lee 2019 publishes only cohort aggregates. No motor-imagery EEG dataset we use publishes race / ethnicity.
- **Multi-seed coverage at the Tier 1+2 layer is now in place.** Experiment 34 provides 5-seed sweeps over experiments 20, 24, and the four cross-dataset directions; the initial A4 PhysioNet result already carries a 5-seed extension via experiment 14. Each Tier 1 headline number lands within ~ 3 pp of its seed-0 value; the Lee 2019 $\rightarrow$ PhysioNet collapse replicates at 0.495 ± 0.006 .

11 Conclusion

We have presented a reproducible, audit-clean benchmark of EEG biometric leakage from motor-imagery BCI models across three independent corpora and four defense families, with the strongest empirical attack we are aware of (DP-aware MIA, encoder fine-tune, Fredrikson model inversion) run against each defense.

The empirical case for treating EEG as biometric data is strong and corroborated on multiple corpora. The empirical defense story is attack-specific and now quantitatively pinned: only DP-SGD $\varepsilon \leq 1$ and federated DP-FedAvg survive the adaptive-attacker stress tests; loose-budget DP-SGD ($\varepsilon \geq 3$) does not protect against MIA when the attacker matches the defense; D1 / D2 collapse above the no-defense baseline. The cross-dataset asymmetry surfaces a substantive scientific finding: a controlled mechanism test (experiment 33) rules out the task-complexity hypothesis and points to recording-rig domain shift as the most plausible remaining mechanism. The Lee 2019 $\rightarrow$ PhysioNet collapse replicates at 0.495 ± 0.006 across five random seeds.

All numbers are reproducible from the canonical JSON files under `results/`, the experiment scripts under `experiments/`, and the run-trail metadata under `runs/`. The current commit returns **273 OK / 0 WARN / 0 FAIL** on the audit.

Data, Code, and Ethics Statement

All experiments use public, de-identified EEG datasets: PhysioNet EEG-MMIDB, BCI Competition IV-2a, and Lee 2019 OpenBMI. The benchmark code, canonical result JSON files, figure-regeneration scripts, and audit trail are organized in the accompanying repository. The submitted

results should be interpreted as a privacy-risk audit of public BCI benchmarks, not as a clinical diagnostic system or a deployed biometric system. Because the experiments use previously released public datasets, no new human-subject recordings were collected for this study. Any future release of larger EEG-biometric benchmarks should include explicit consent language for identity-linkage and membership-inference risk, along with demographic metadata sufficient for subgroup privacy audits.

Author Contributions

Manraj Singh Mondair designed the benchmark, implemented the attack and defense pipelines, ran the experiments, audited the results, and wrote the manuscript.

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